

CEIS101

Smart Home Automation and Security System

CEIS101 Final Course Project

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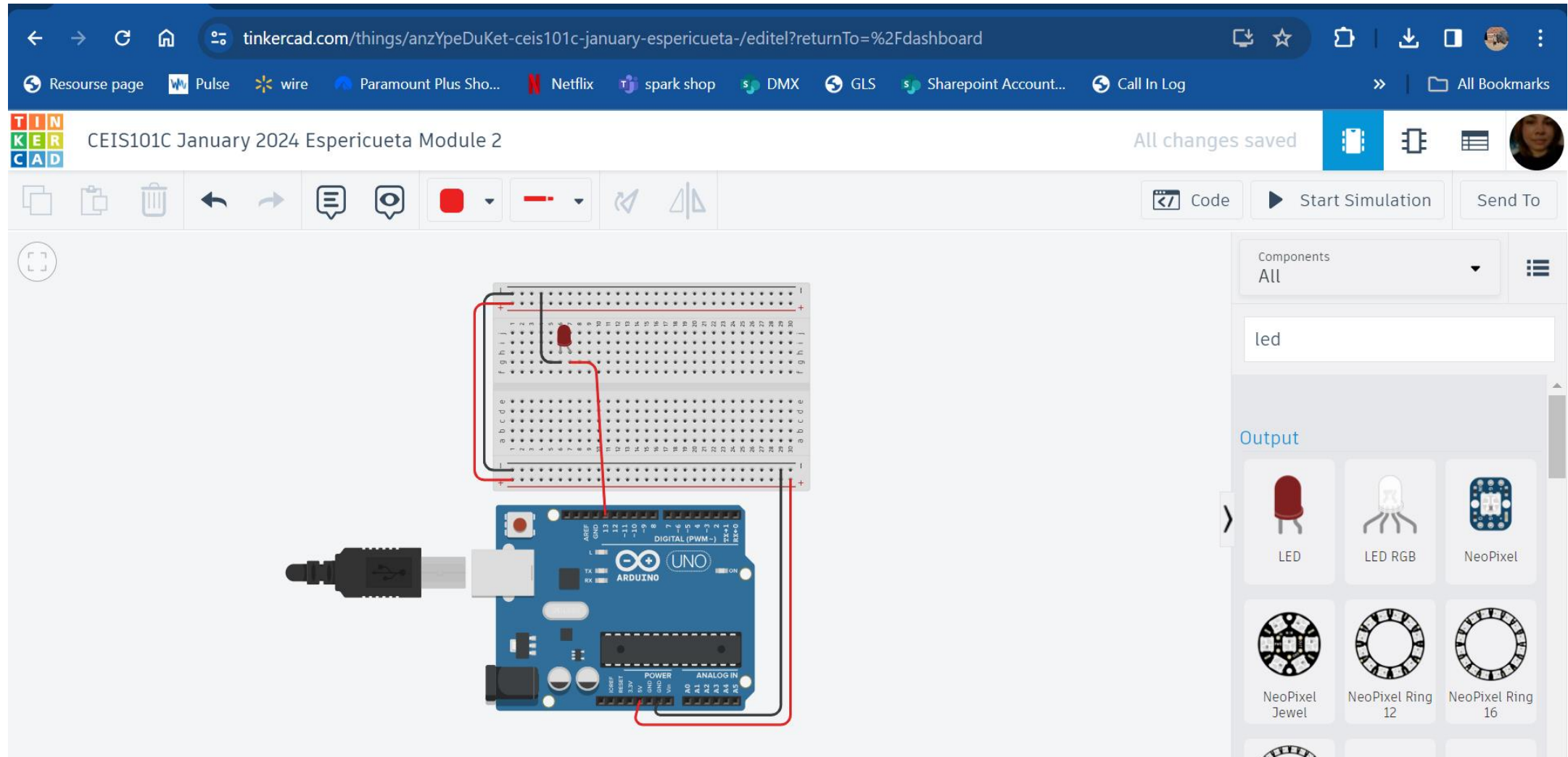
Introduction

Smart Home Automated Security System:

- Overall Project: Simulating and building a circuit in Tinkercad. Adding a door sensor to the smart home security system to check whether the door is open or closed. Adding a distance sensor to the smart home security system to monitor for a possible home intruder; and adding an automated light to the smart home.
- Objective: The obtain the fundamental concept of the IoT by integrating hardware, software, and networks into a whole system.
- Tools: Tinkercad, Microsoft Word, Microsoft Excel, PowerPoint, and Wix

Circuit Simulation in Tinkercad

Circuit (screenshot)



Code (screenshot)

The screenshot displays the Tinkercad web application interface. At the top, the browser address bar shows the URL: `tinkercad.com/things/anzYpeDuKet-ceis101c-january-espericueta-/editel?returnTo=%2Fdashboard`. Below the address bar, a navigation bar includes links for Resource page, Pulse, wire, Paramount Plus Sho..., Netflix, spark shop, DMX, GLS, Sharepoint Account..., and Call In Log. The main workspace is titled "CEIS101C January 2024 Espericueta Module 2" and shows "All changes saved". The workspace is divided into three main sections: a circuit diagram on the left, a block-based code editor in the center, and a text-based code editor on the right.

The circuit diagram on the left shows an Arduino Uno R3 board connected to a breadboard. A red wire is connected to pin 13, and a black wire is connected to ground. The block-based code editor in the center contains the following sequence of blocks:

- on start
- set pin 13 to HIGH
- wait 1 secs
- set pin 13 to LOW
- wait 1 secs
- repeat 10 times

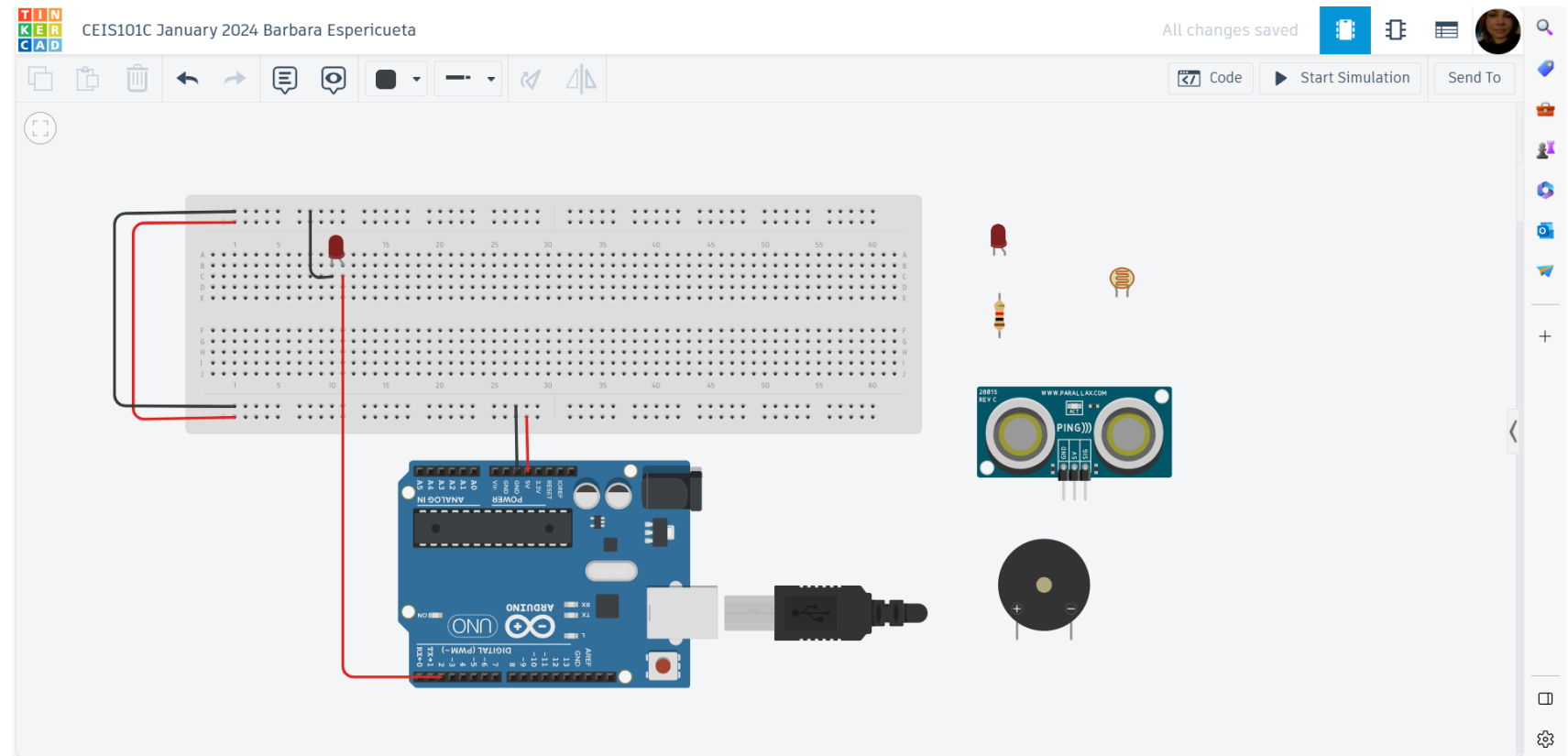
The text-based code editor on the right shows the corresponding C++ code:

```
1 // C++ code
2 //
3 void setup()
4 {
5   pinMode(13, OUTPUT);
6 }
7
8 void loop()
9 {
10  digitalWrite(13, HIGH);
11  delay(500); // Wait for 1000 millisecond(s)
12  digitalWrite(13, LOW);
13  delay(1000); // Wait for 1000 millisecond(s)
14 }
```

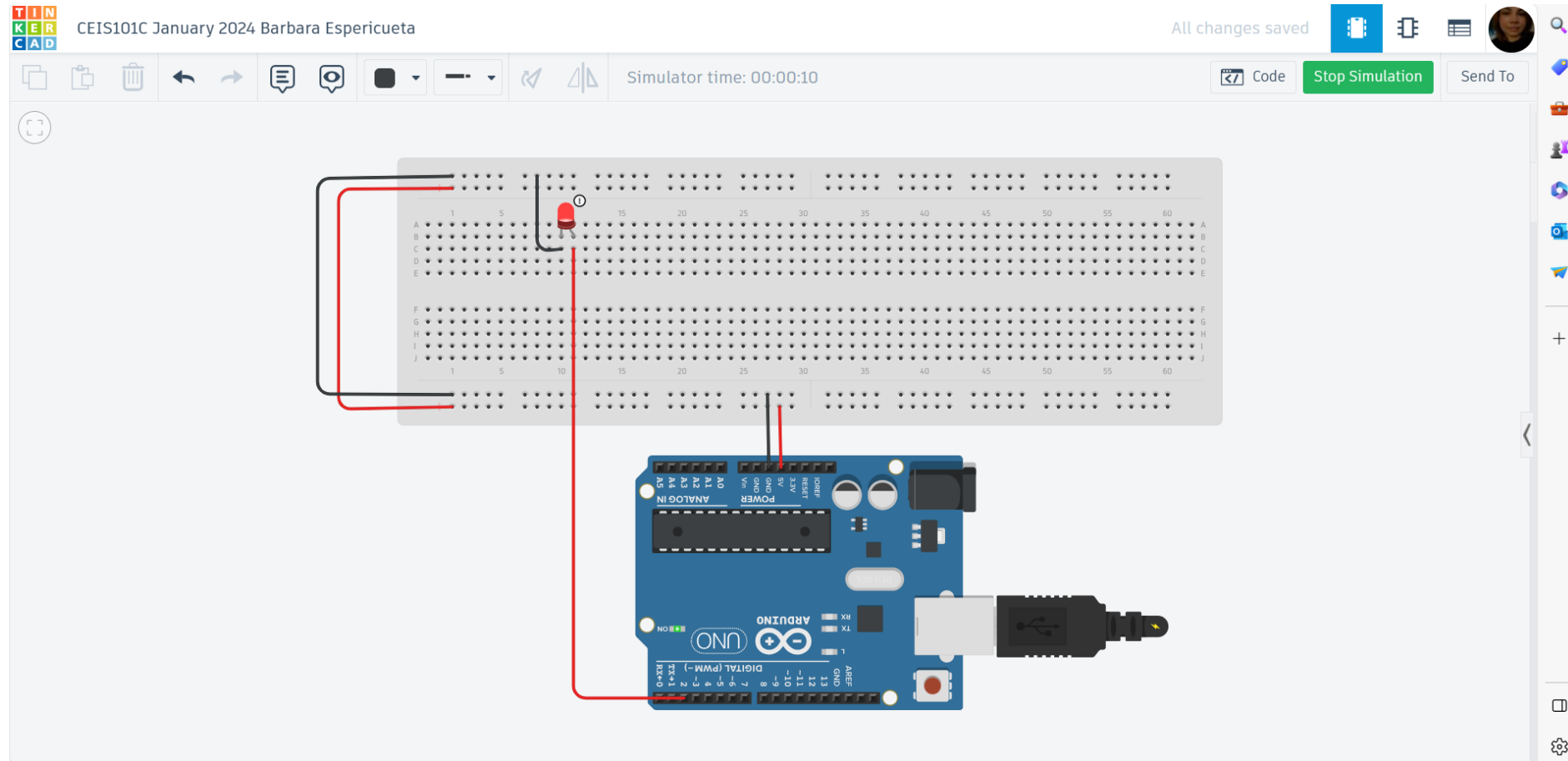
Inventory of Parts, Circuit Building, and Displaying Messages

Organization of Project Components (picture)

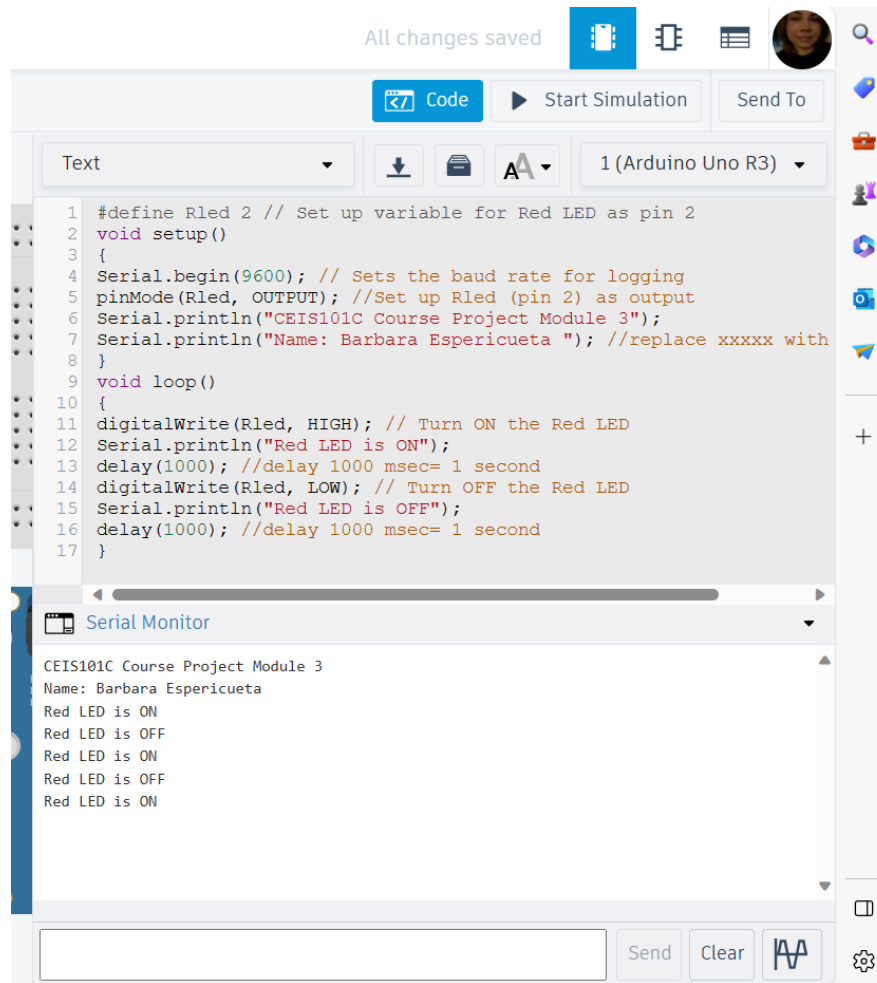
- Arduino Uno
- Breadboard
- Resistor 10k Ω
- LEDs
- Ultrasonic Sensor
- Active Buzzer
- Photoresistor
- Wires



Circuit with red LED on (picture)

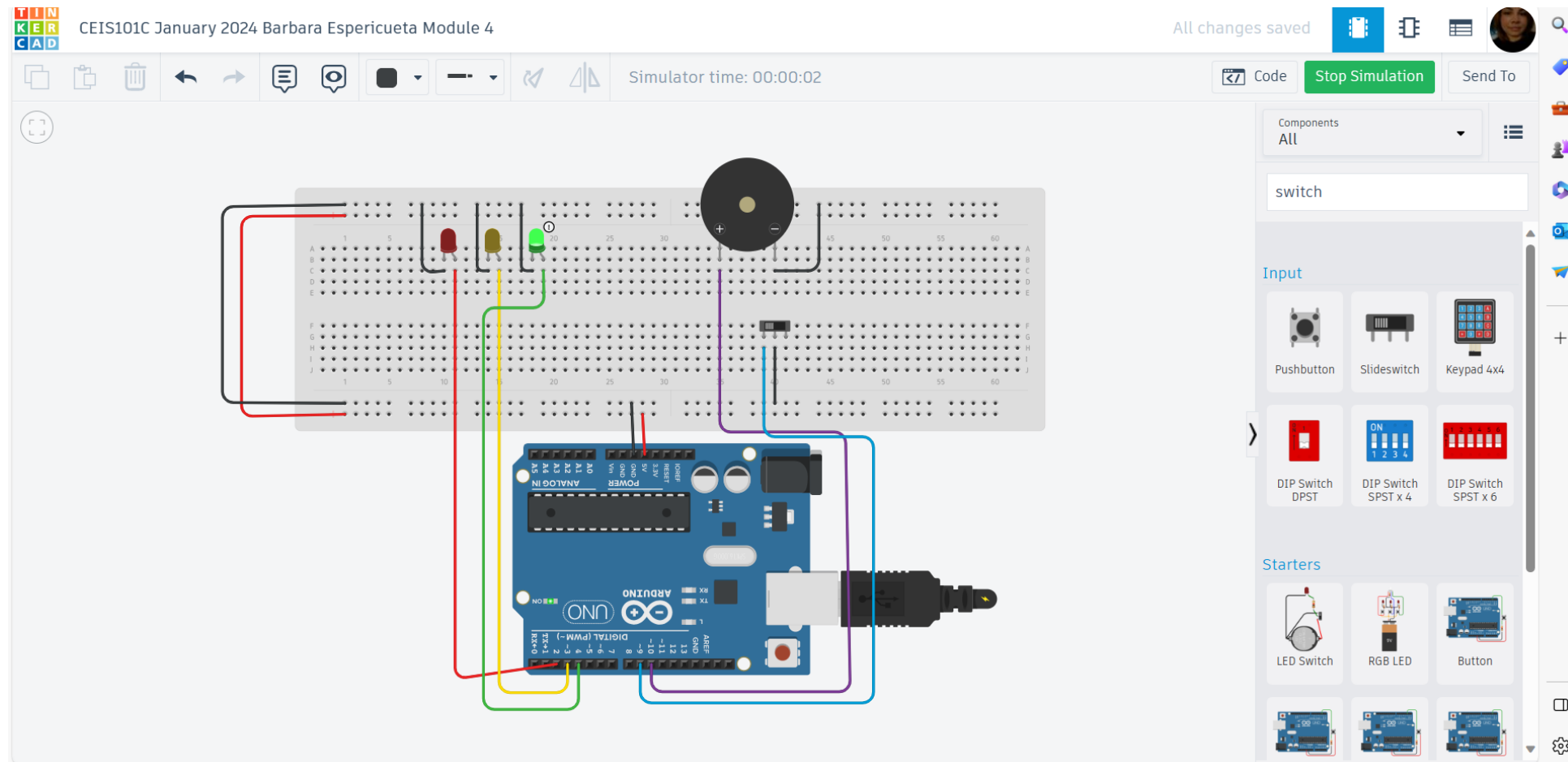


Serial Monitor (screenshot)

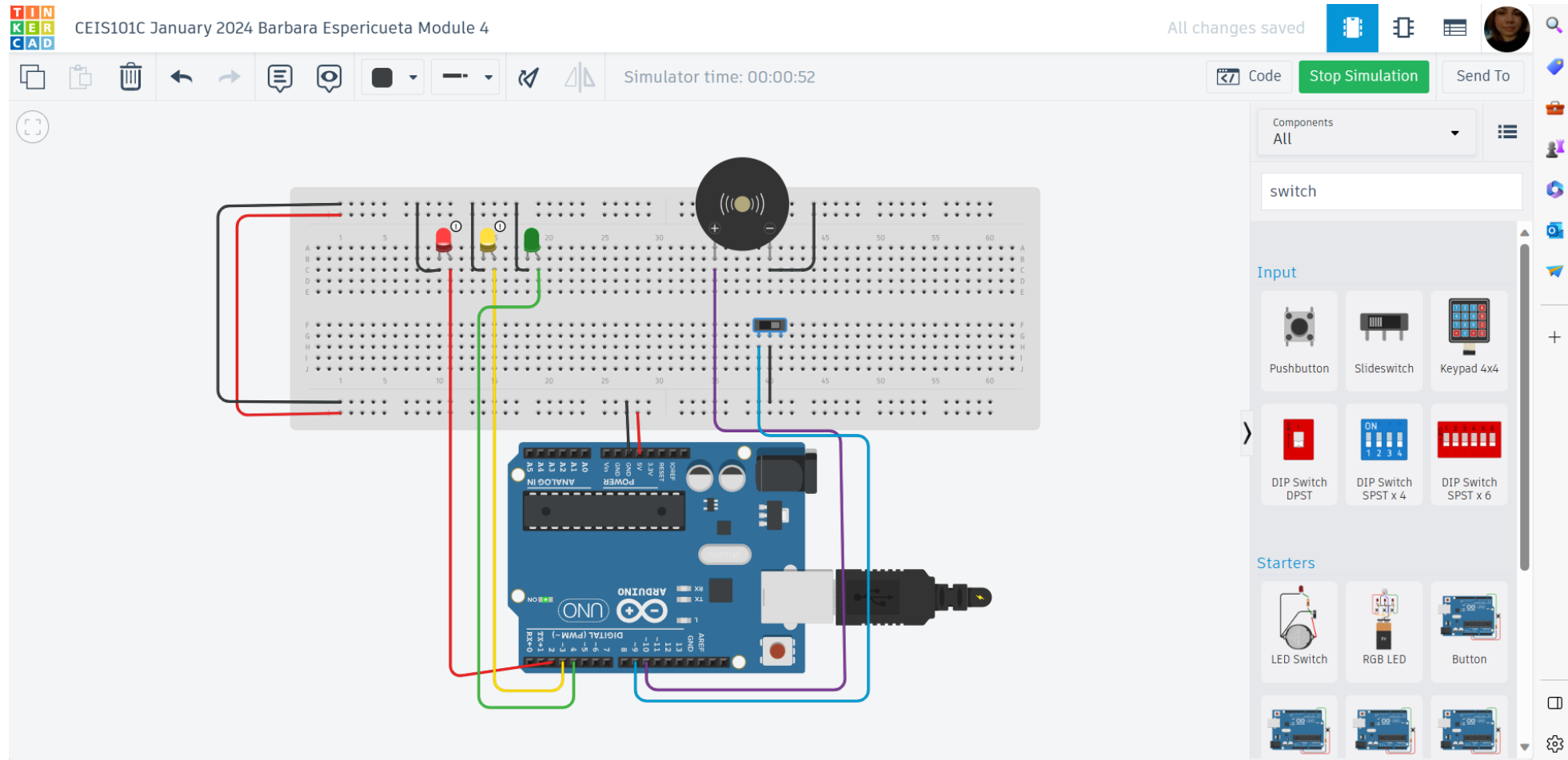


Adding Door Sensor to Smart Home System

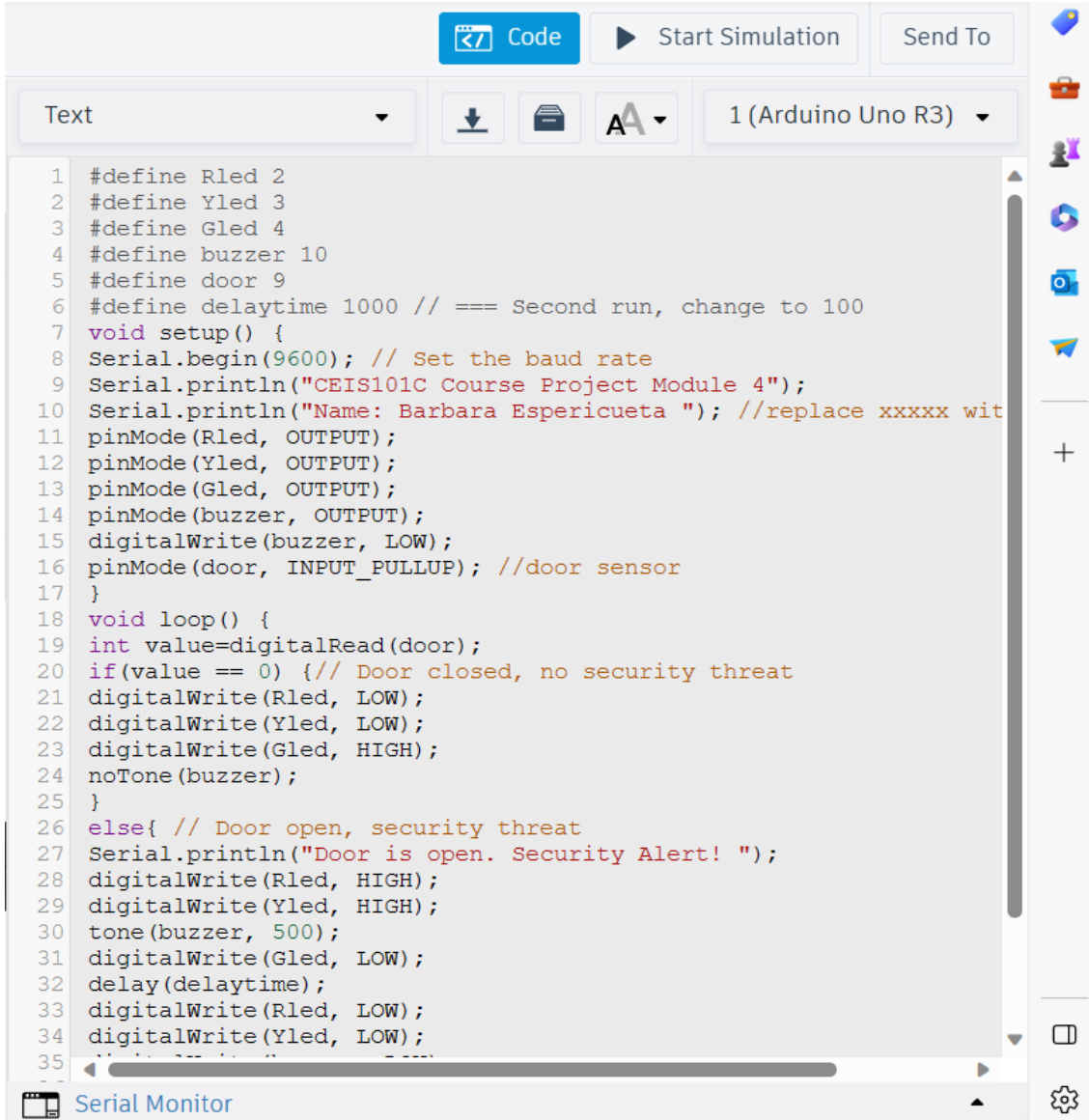
Circuit of door closed with Green LED ON (picture)



Circuit of door open with Green LED OFF (picture)



Arduino Code (screenshot)

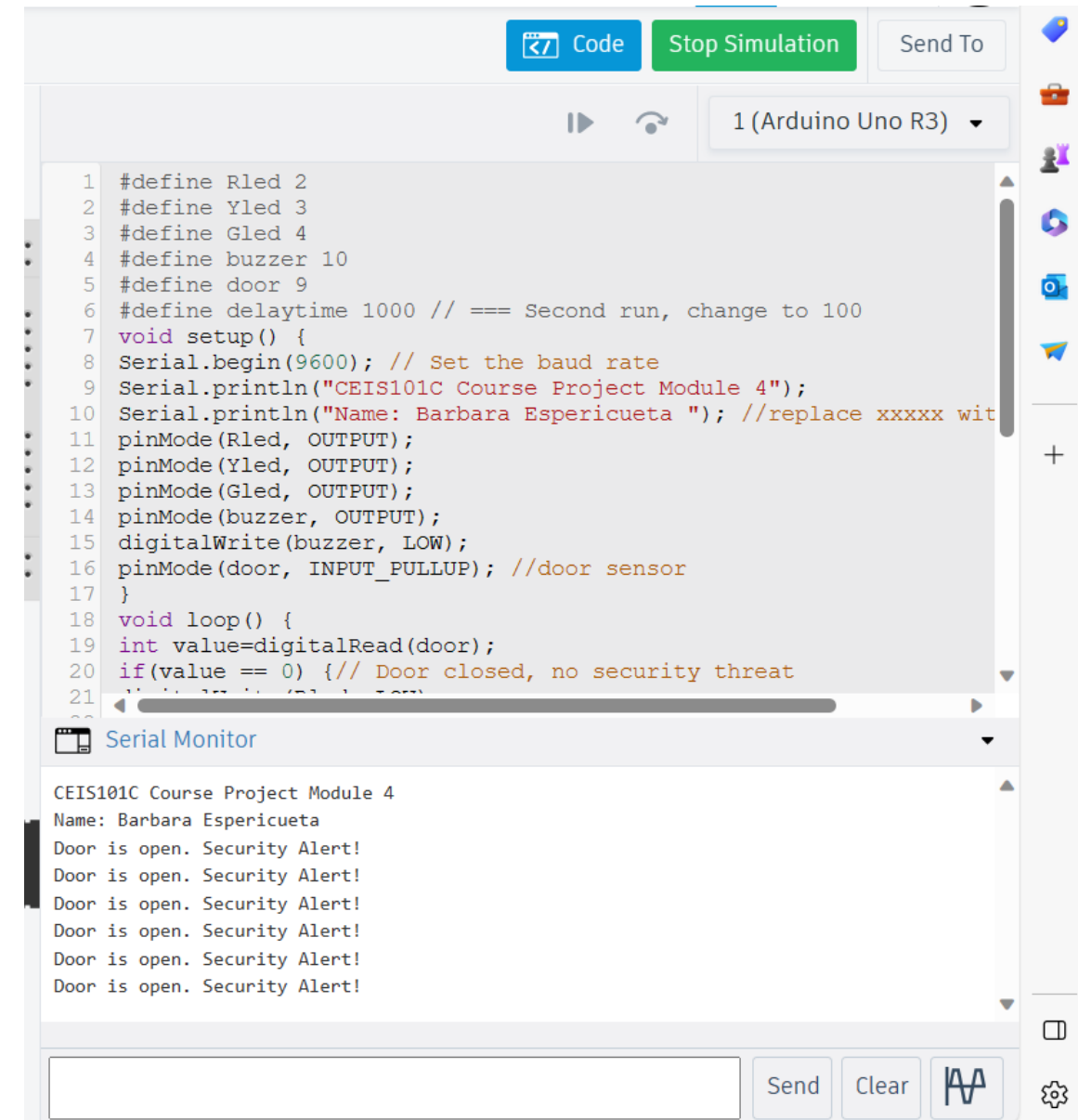


```

1 #define Rled 2
2 #define Yled 3
3 #define Gled 4
4 #define buzzer 10
5 #define door 9
6 #define delaytime 1000 // === Second run, change to 100
7 void setup() {
8   Serial.begin(9600); // Set the baud rate
9   Serial.println("CEIS101C Course Project Module 4");
10  Serial.println("Name: Barbara Espericueta "); //replace xxxxx wit
11  pinMode(Rled, OUTPUT);
12  pinMode(Yled, OUTPUT);
13  pinMode(Gled, OUTPUT);
14  pinMode(buzzer, OUTPUT);
15  digitalWrite(buzzer, LOW);
16  pinMode(door, INPUT_PULLUP); //door sensor
17 }
18 void loop() {
19   int value=digitalRead(door);
20   if(value == 0) { // Door closed, no security threat
21     digitalWrite(Rled, LOW);
22     digitalWrite(Yled, LOW);
23     digitalWrite(Gled, HIGH);
24     noTone(buzzer);
25   }
26   else{ // Door open, security threat
27     Serial.println("Door is open. Security Alert! ");
28     digitalWrite(Rled, HIGH);
29     digitalWrite(Yled, HIGH);
30     tone(buzzer, 500);
31     digitalWrite(Gled, LOW);
32     delay(delaytime);
33     digitalWrite(Rled, LOW);
34     digitalWrite(Yled, LOW);
35

```

Serial Monitor (screenshot)



The screenshot displays the Arduino IDE interface. The top toolbar includes buttons for 'Code', 'Stop Simulation', and 'Send To'. Below the toolbar, a dropdown menu shows '1 (Arduino Uno R3)'. The main text area contains the following C++ code:

```
1 #define Rled 2
2 #define Yled 3
3 #define Gled 4
4 #define buzzer 10
5 #define door 9
6 #define delaytime 1000 // == Second run, change to 100
7 void setup() {
8   Serial.begin(9600); // Set the baud rate
9   Serial.println("CEIS101C Course Project Module 4");
10  Serial.println("Name: Barbara Espericueta "); //replace xxxxx with
11  pinMode(Rled, OUTPUT);
12  pinMode(Yled, OUTPUT);
13  pinMode(Gled, OUTPUT);
14  pinMode(buzzer, OUTPUT);
15  digitalWrite(buzzer, LOW);
16  pinMode(door, INPUT_PULLUP); //door sensor
17 }
18 void loop() {
19   int value=digitalRead(door);
20   if(value == 0) { // Door closed, no security threat
21     // ... (code is partially obscured)
22   }
```

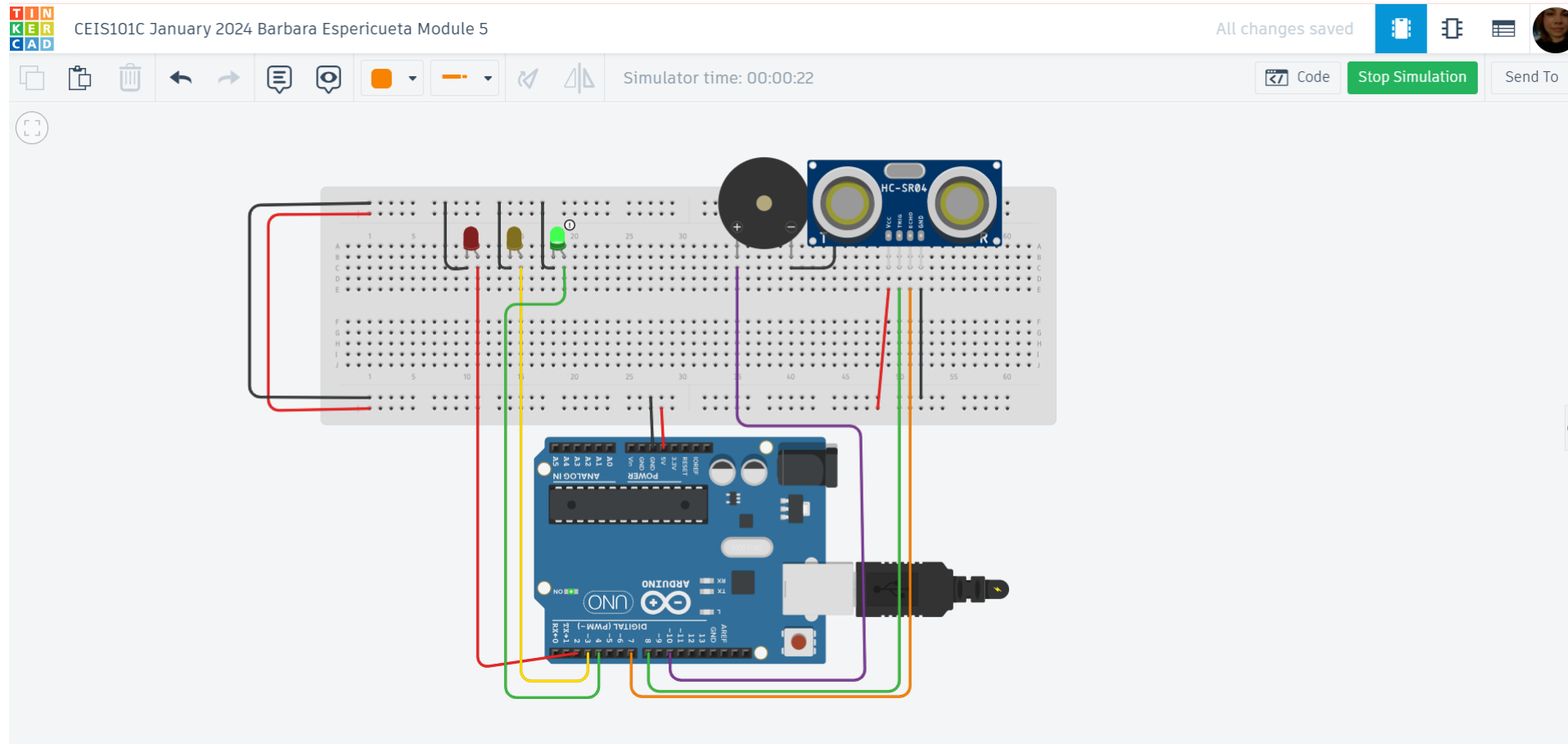
Below the code editor, the 'Serial Monitor' window is open, showing the output of the program:

```
CEIS101C Course Project Module 4
Name: Barbara Espericueta
Door is open. Security Alert!
Door is open. Security Alert!
Door is open. Security Alert!
Door is open. Security Alert!
Door is open. Security Alert!
Door is open. Security Alert!
```

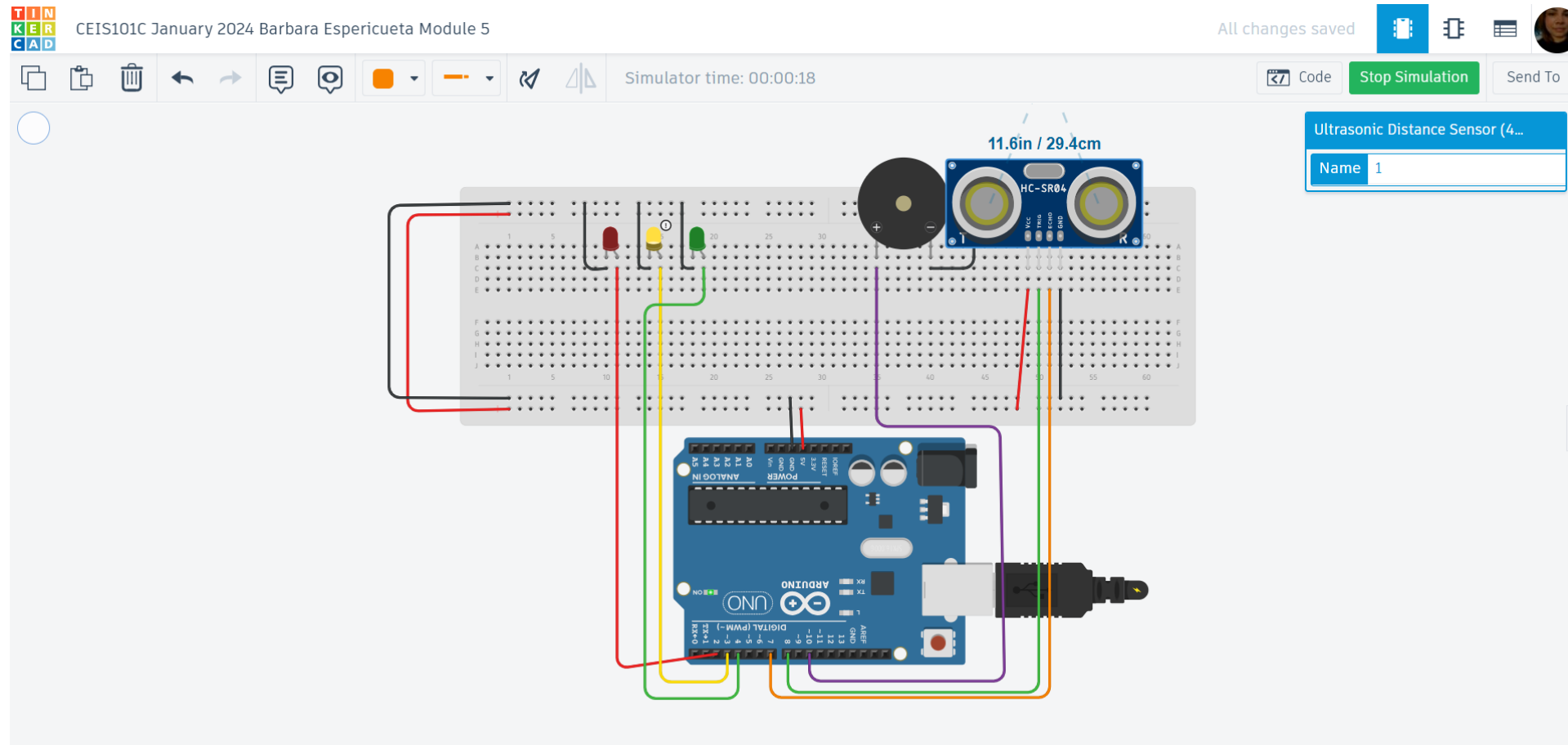
At the bottom of the Serial Monitor window, there is a text input field and buttons for 'Send', 'Clear', and a waveform icon.

Adding Distance Sensor to Smart Home System and Conducting Data Analysis

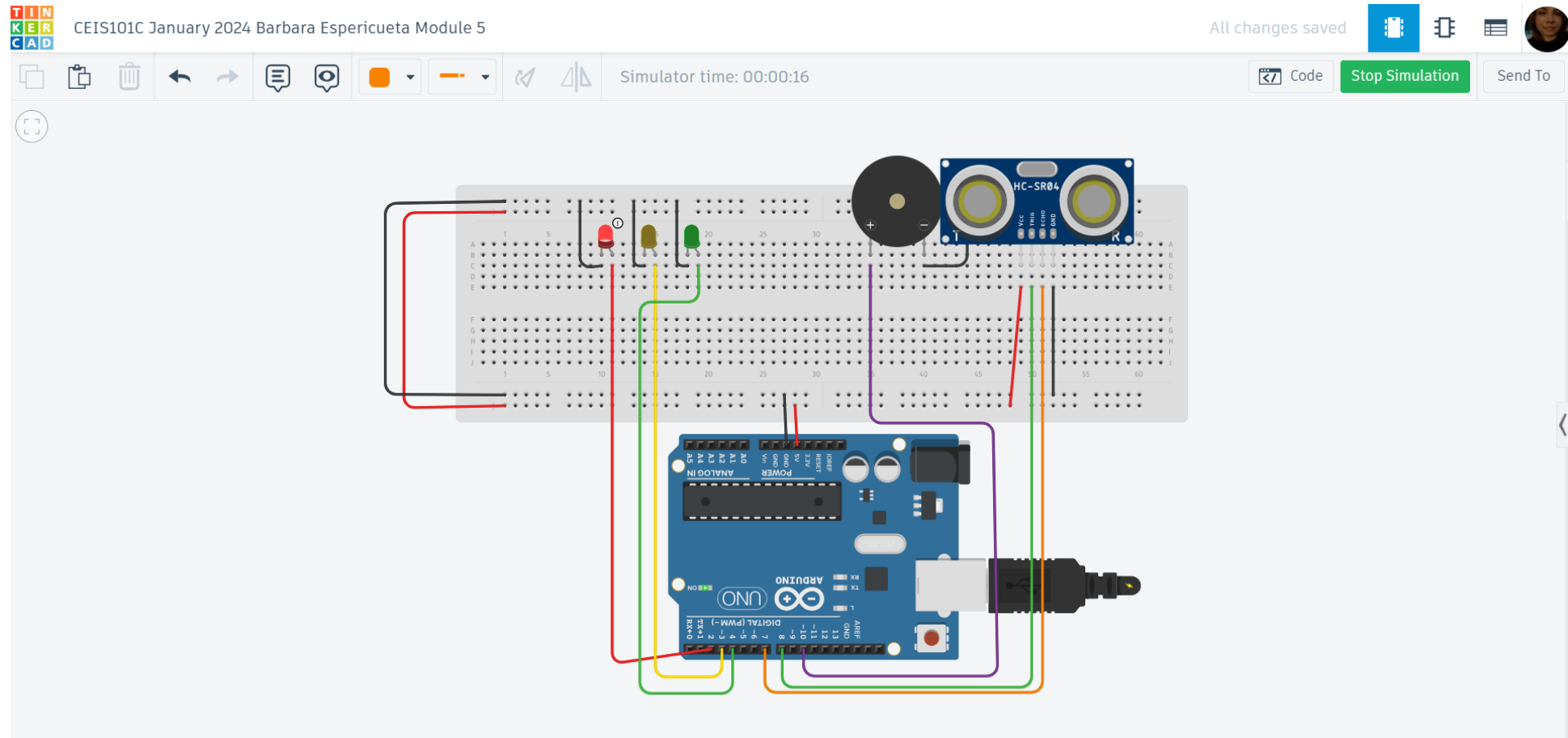
Circuit with green LED on (picture)



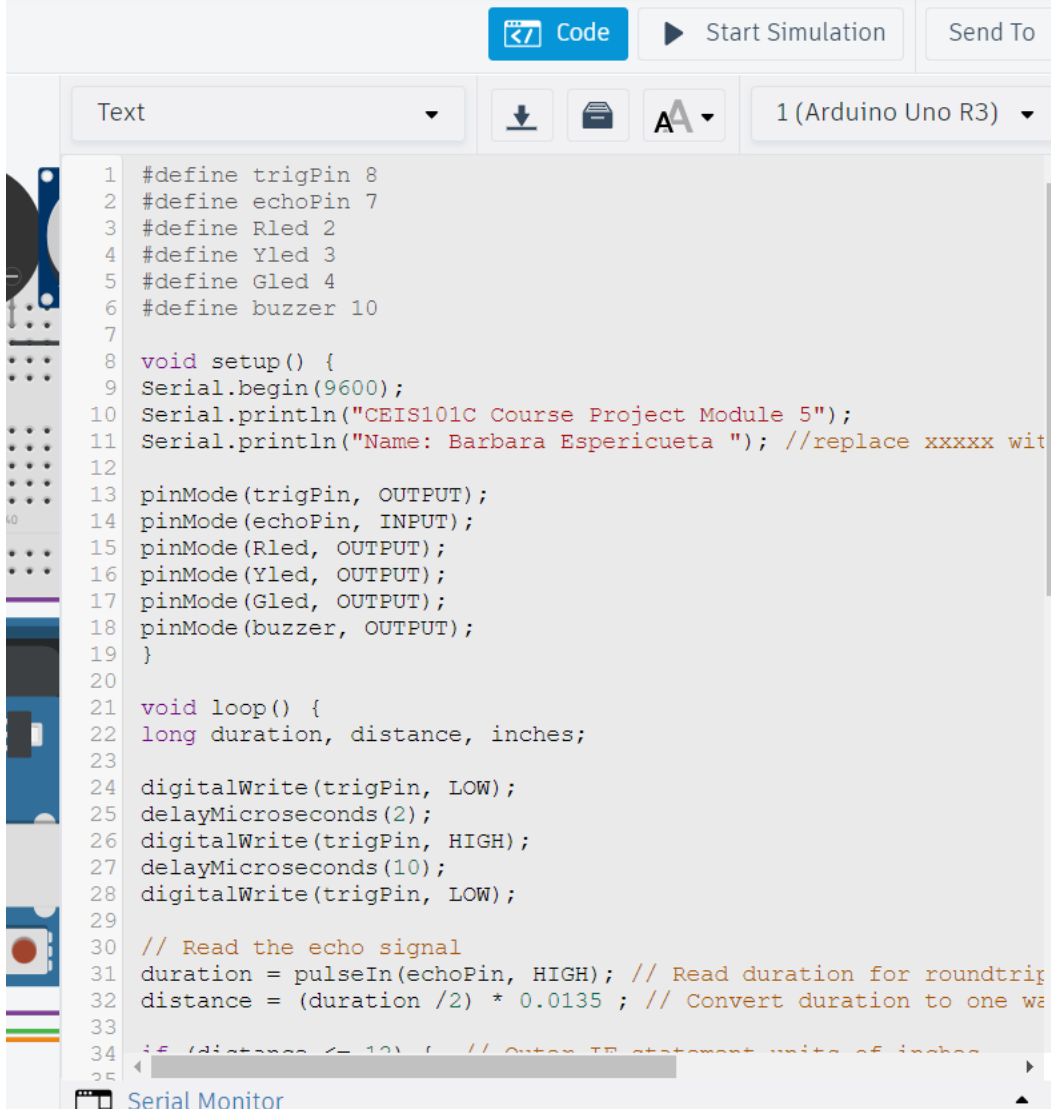
Circuit with yellow LED on (picture)



Circuit with red LED on (picture)



Arduino Code (screenshot)



The screenshot shows the Arduino IDE interface with a C++ code file open. The code is for an Arduino Uno R3 and defines pins for a trigPin, echoPin, and several LEDs and a buzzer. It includes setup and loop functions for serial communication and distance measurement using a pulseIn function.

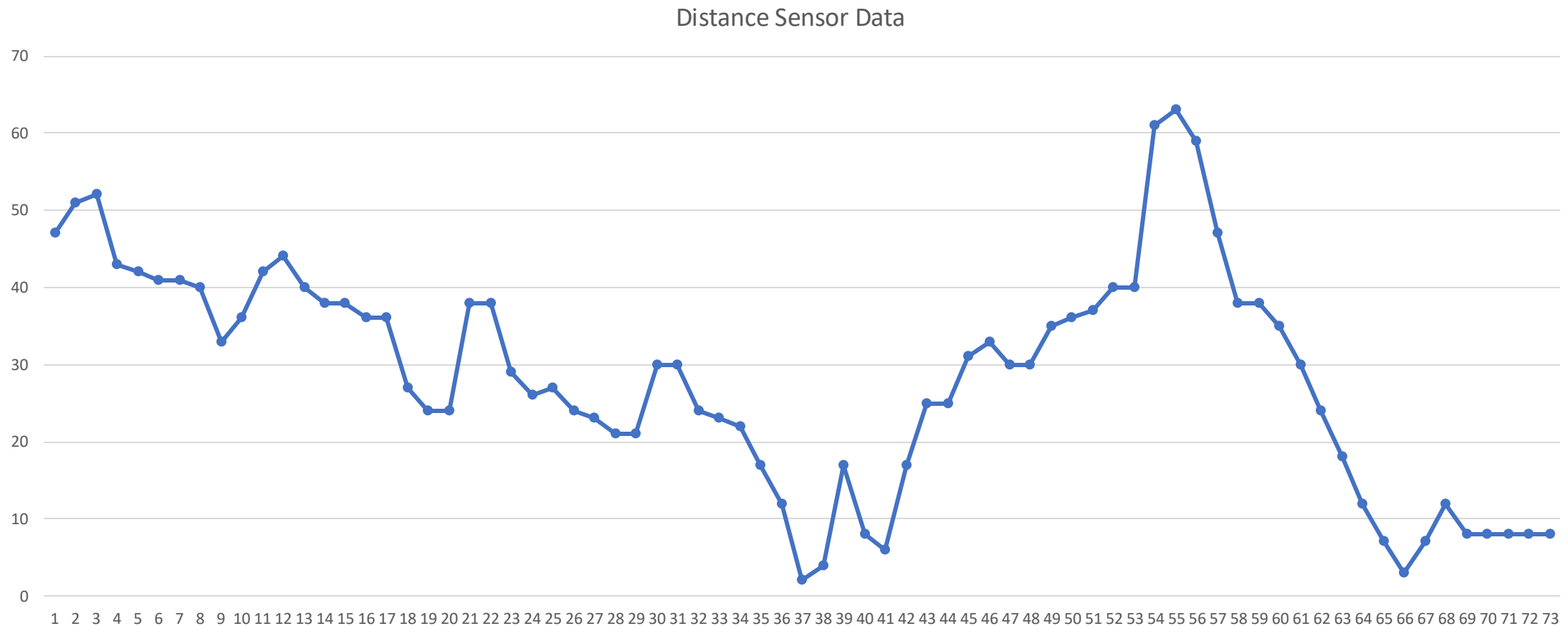
```

1  #define trigPin 8
2  #define echoPin 7
3  #define Rled 2
4  #define Yled 3
5  #define Gled 4
6  #define buzzer 10
7
8  void setup() {
9    Serial.begin(9600);
10   Serial.println("CEIS101C Course Project Module 5");
11   Serial.println("Name: Barbara Espericueta "); //replace xxxxx wit
12
13   pinMode(trigPin, OUTPUT);
14   pinMode(echoPin, INPUT);
15   pinMode(Rled, OUTPUT);
16   pinMode(Yled, OUTPUT);
17   pinMode(Gled, OUTPUT);
18   pinMode(buzzer, OUTPUT);
19 }
20
21 void loop() {
22   long duration, distance, inches;
23
24   digitalWrite(trigPin, LOW);
25   delayMicroseconds(2);
26   digitalWrite(trigPin, HIGH);
27   delayMicroseconds(10);
28   digitalWrite(trigPin, LOW);
29
30   // Read the echo signal
31   duration = pulseIn(echoPin, HIGH); // Read duration for roundtrip
32   distance = (duration / 2) * 0.0135 ; // Convert duration to one wa
33
34   if (distance <= 12) { // Output IF statement units of inches
35

```

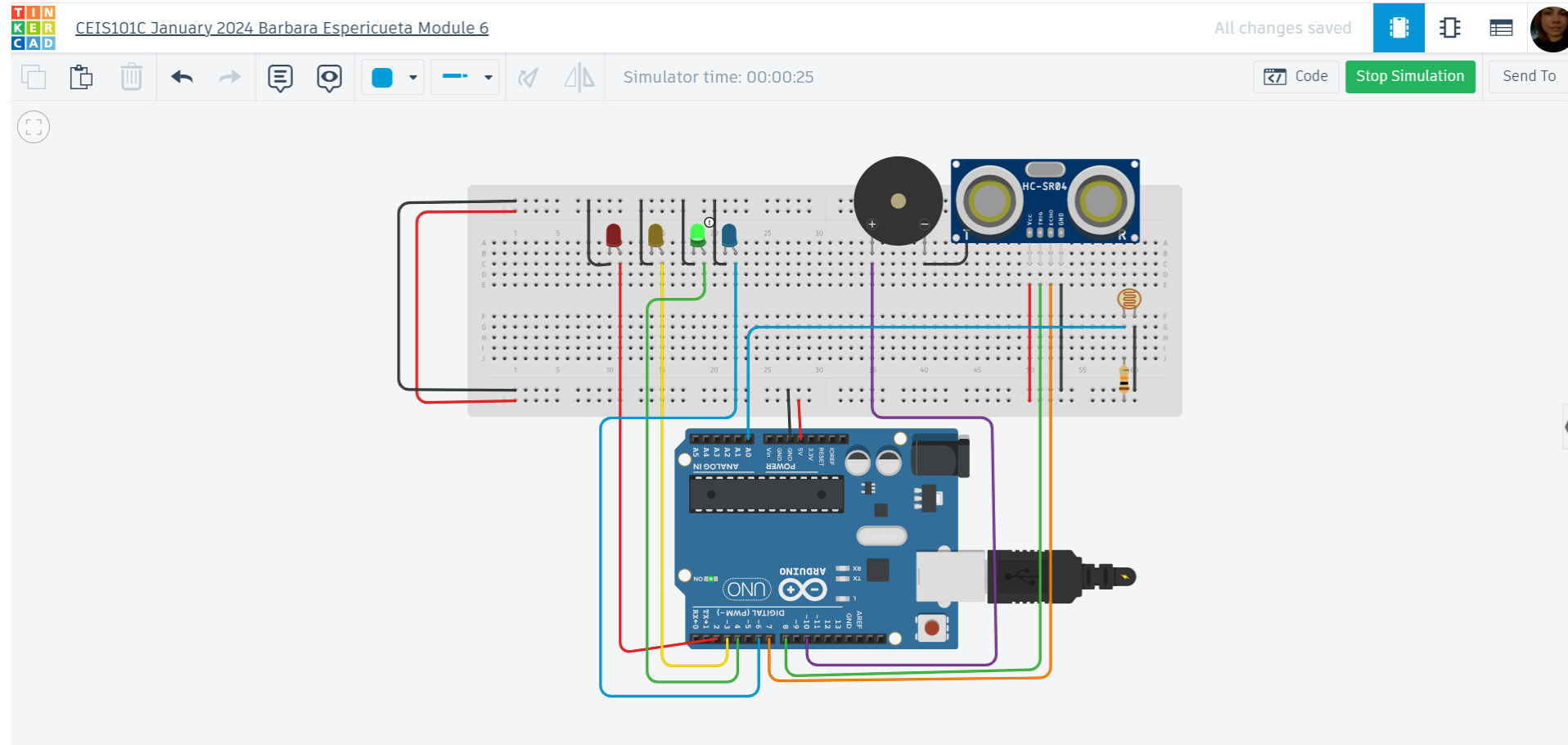
Serial Monitor

Plot of data (graph from Excel)

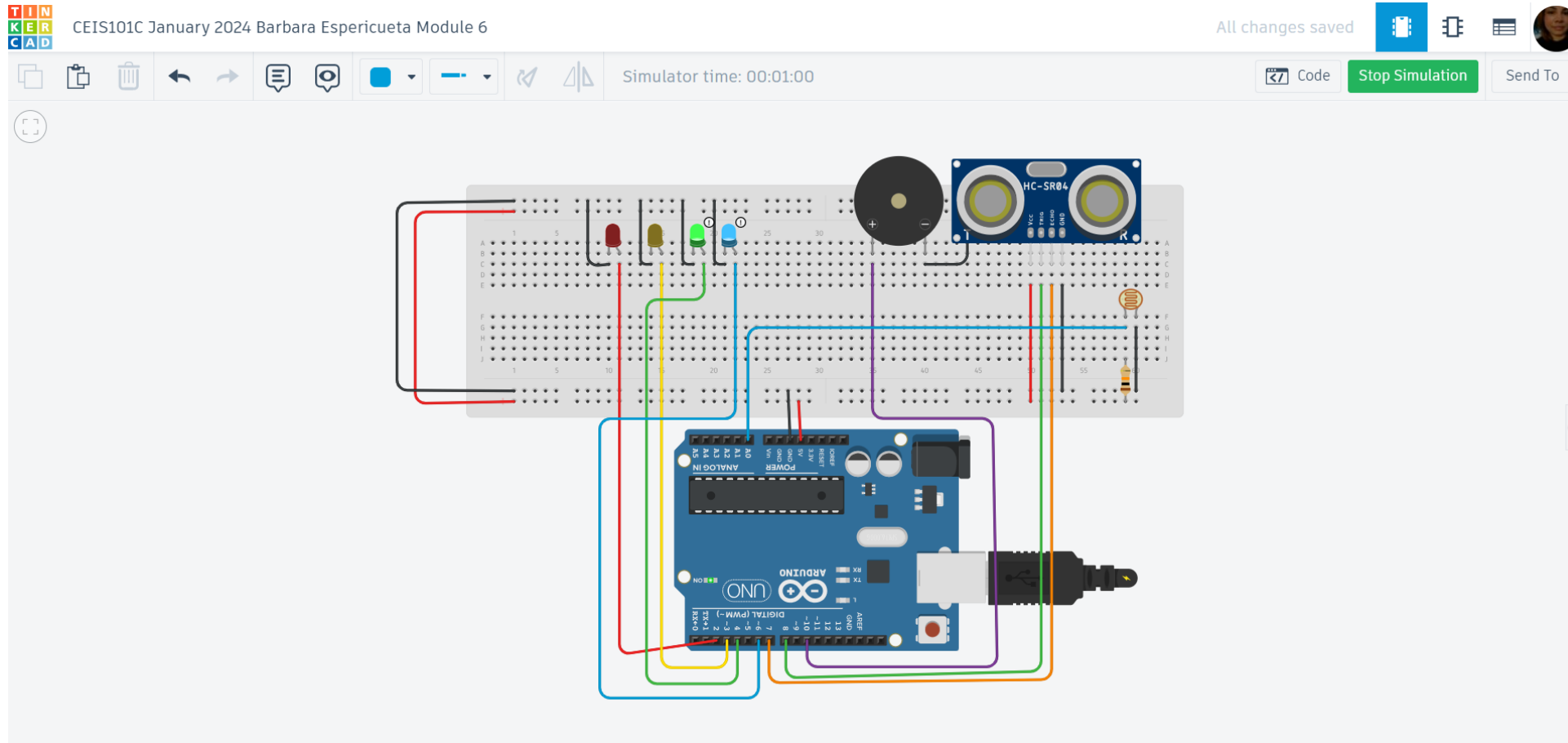


Adding Automated Light to Smart Home System

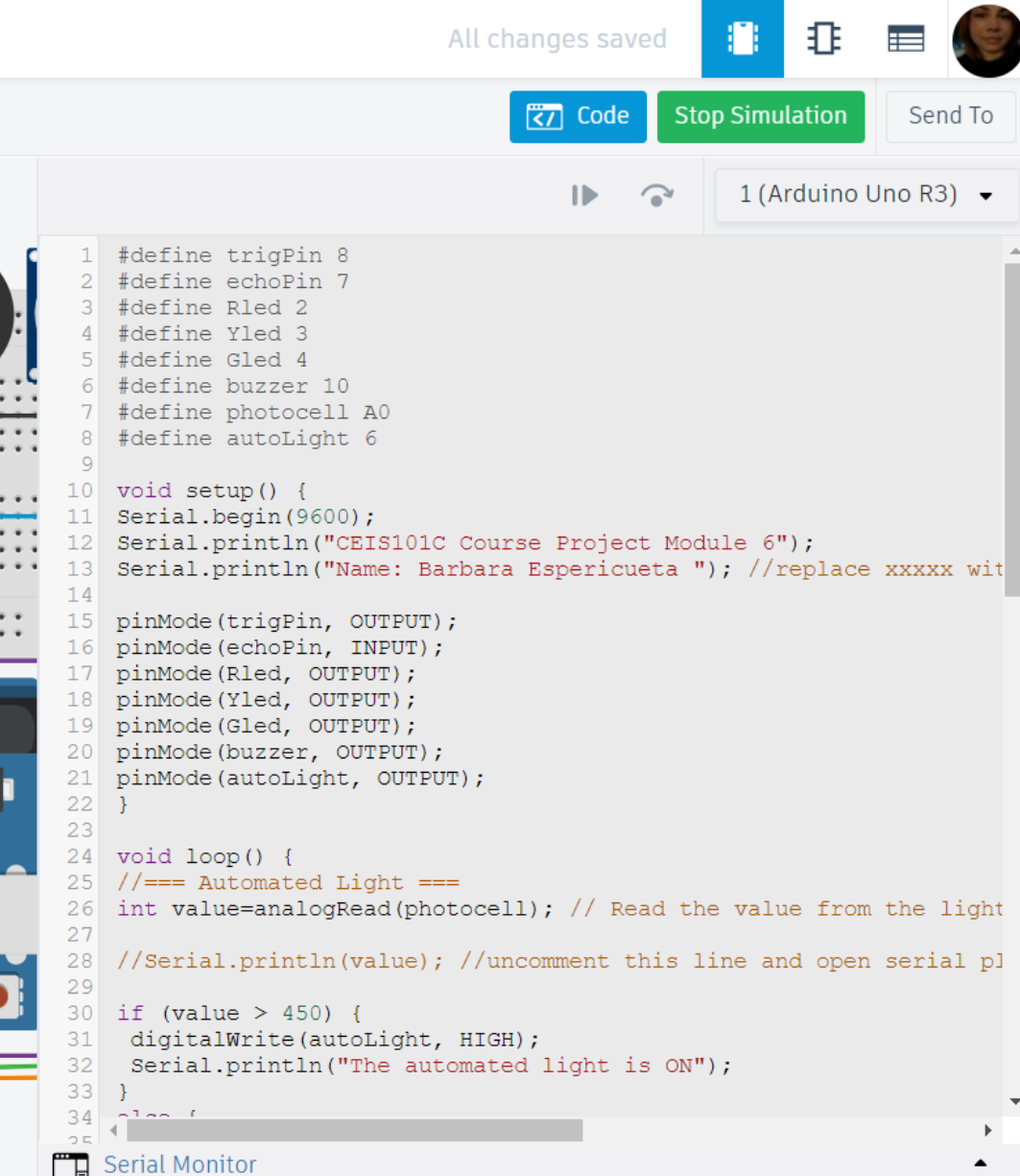
Circuit with automated LED off (picture)



Circuit with automated LED on (picture)



Arduino Code (screenshot)



The screenshot shows the Arduino IDE interface. At the top, there's a header with the DeVry University logo and name. Below it, a status bar indicates "All changes saved". On the right, there are icons for a document, a circuit board, and a list. A user profile picture is also visible. Below the header, there are buttons for "Code", "Stop Simulation", and "Send To". A dropdown menu shows "1 (Arduino Uno R3)". The main area displays the following code:

```

1  #define trigPin 8
2  #define echoPin 7
3  #define Rled 2
4  #define Yled 3
5  #define Gled 4
6  #define buzzer 10
7  #define photocell A0
8  #define autoLight 6
9
10 void setup() {
11   Serial.begin(9600);
12   Serial.println("CEIS101C Course Project Module 6");
13   Serial.println("Name: Barbara Espericueta "); //replace xxxxx with
14
15   pinMode(trigPin, OUTPUT);
16   pinMode(echoPin, INPUT);
17   pinMode(Rled, OUTPUT);
18   pinMode(Yled, OUTPUT);
19   pinMode(Gled, OUTPUT);
20   pinMode(buzzer, OUTPUT);
21   pinMode(autoLight, OUTPUT);
22 }
23
24 void loop() {
25   //=== Automated Light ===
26   int value=analogRead(photocell); // Read the value from the light
27
28   //Serial.println(value); //uncomment this line and open serial p
29
30   if (value > 450) {
31     digitalWrite(autoLight, HIGH);
32     Serial.println("The automated light is ON");
33   }
34   delay(1000);
35 }

```

At the bottom, there is a "Serial Monitor" tab.

Serial Monitor (screenshot)

All changes saved    

 Code  Start Simulation  Send To

Text    1 (Arduino Uno R3) 

```

1  #define trigPin 8
2  #define echoPin 7
3  #define Rled 2
4  #define Yled 3
5  #define Gled 4
6  #define buzzer 10
7  #define photocell A0
8  #define autoLight 6
9
10 void setup() {
11   Serial.begin(9600);
12   Serial.println("CEIS101C Course Project Module 6");
13   Serial.println("Name: Barbara Espericueta "); //replace xxxxxx with
14
15

```

 Serial Monitor

```

CEIS101C Course Project Module 6
Name: Barbara Espericueta
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
The automated light is ON
CEIS101C Course Project Module 6
Name: Barbara Espericueta
The automated light is ON
The automated light is ON
The automated light is ON

```

 Send  Clear 

Challenges/Lessons Learned

- I had challenges explaining what I've learned. Using the right terminology and information to explain what I understood about the lessons.
- Staying persistent and taking the time to read the lessons. Really diving into the discussions by researching and studying from many different platforms. Getting different perspectives from peers was a plus, helping me overcome my challenges.
- Lessons I learned during this course were to stay disciplined, stay persistent, and manage stress. This will help you stay on top of your learning.

Career Skills

- Skills I developed are network awareness, performance of the prototype, different programming languages, computer language, security awareness, and professionalism.
- What competencies will help you to gain the opportunities to advance your career? Professionalism, employers will know that I am a professional in my skill, that I am knowledgeable and will complete with proficiency and with integrity.

Conclusion

I have successfully completed in simulating and building a circuit in Tinkercad. Obtain the fundamental concept of the IoT. Gaining skills in Network awareness, performance of a prototype, computer language, and security awareness. All in learning to stay disciplined, staying persistent, and staying professional.